

Cleartrip: Improving an end-to-end travel experience with a high- performance infrastructure

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About Cleartrip

Cleartrip helps customers plan their ideal travel journey by offering flights, hotel, and train options. Cleartrip provides a seamless booking experience through its platform, supported by BigQuery and Google Kubernetes Engine.

Cleartrip is the leading online travel and leisure aggregator in the Middle East and India.

Founded in 2006, Cleartrip offers its customers a comprehensive and personalized travel experience through its innovative and award-winning mobile and desktop solutions.

Combining intuitive products with a customer-centric approach and the widest selection of flights and hotels, Cleartrip brings a highly differentiated value proposition to market, offering convenience, choice, competitive prices, and exclusive content.

Google Cloud results

- Saves 10% of operational costs and improves uptime for a more reliable app
- Empowers engineers to take ownership by giving them full data visibility and control
- Monitoring app performance automatically with Cloud Monitoring to improve customer experience
- Easily manages traffic spikes during festivities with the containerization of Google Kubernetes Engine

Help
90%
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issu
Goo

Industries: Travel & Hospitality

Location: India

(<https://cloud.google.com/monitoring/>), [Cloud Google Cloud \(https://cloud.google.com/\)](https://cloud.google.com/)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

BigQuery (<https://cloud.google.com/bigquery/>)
(<https://cloud.google.com/storage/>)

Monitoring (<https://cloud.google.com/monitoring/>)

Cloud Memorystore
(<https://cloud.google.com/memorystore/>)

Operations
(<https://cloud.google.com/products/operations>)

Cloud Storage (<https://cloud.google.com/storage/>)

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—*Manoj Sharma, CTO,
Cleartrip*

Travel should be an enjoyable experience, which is why having a travel platform that simplifies bookings

and condenses everything travelers need in one place is handy. This is exactly what Cleartrip does, through its single application that is accessible on mobile and desktop. From planning an itinerary to booking flights, trains, and hotels, customers can navigate their travel journey with ease from the moment they decide to make a trip.

"We are in the business of offering travel solutions to our customers, so having zero downtime is essential. For this reason, we believe that Google Cloud is a very viable solution for us because of its high availability and scalability." The company began its migration to [Google Cloud](https://cloud.google.com/) (<https://cloud.google.com/>) in August 2019. Since then, it has solved up to 90% of production issues which include network and resources utilization.

Needless to say, cost is a big factor when running a business, and upon doing its due diligence in calculating costs, Cleartrip found that Google Cloud was the most viable option. "By moving our production to Google Cloud and keeping dev-test and quality on-prem on old hardware, we found that we saved 10% of our operational costs, with improved uptime."

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Moving to cloud for transparency and better visibility

As a customer-facing application, Cleartrip focuses heavily on user experience. "Our focus has always been on the application, so we would only do the minimum required on the infrastructure side to ensure that our applications can run smoothly," says Manoj Sharma, CTO at Cleartrip. As a result, there are many blind spots around data capacity that the application team is unaware of. Such as bottlenecks within the system that hamper the overall speed of processes, which would remain unaddressed unless

someone from the infrastructure team points out the challenges on the data layers.

The transition to Google Cloud was a revelation for Cleartrip because it helped surface many under-resourced data layers which were not maximized.

Manoj says that after moving to Google Cloud, Cleartrip was able to immediately ramp up the capacity of its business by addressing the details that were previously unseen. "Migrating to Google Cloud was an architectural evolution for us. Google Cloud, with Cloud Monitoring (/monitoring) to monitor and troubleshoot application performance, has provided us with significant reliability."

To illustrate how the business has changed since migrating to Google Cloud, Manoj explains that the company usually caches the data it gets from airlines for only a short duration by under-resourcing one Redis. However, in doing so, it was using a higher utilization of the rest of its infrastructure because it required more compute units and network calls to be made to suppliers. This meant that it was processing a lot more data than it had to, and was slow to consumers because it was evicting data from cache. Moving to Memorystore (/memorystore) reduced this latency and ensured high availability, with better visibility and more flexibility to play around with data capacity.

This move excited the engineers at Cleartrip, because having full visibility and control of data capacity on Google Cloud makes launching new initiatives and products like new travel formats features on the platform a breeze. This is a stark

difference to how things were in the past, where the infrastructure team had to manually get hold of additional hardware to set up new infrastructure capacities. When they ran into resource limitations, the team would then have to wait for additional capacity, delaying the process.

The migration process also showed Cleartrip that its systems were capable of working independently from each other. "As we started digging deeper, we came to a realization that our systems could work independently with minor architecture changes, so we were able to remove unnecessary couplings within the systems, which we had initially believed were not possible."

Scaling up quickly with more accurate targeting

Already a leading travel portal in India and the Middle East, Cleartrip has big plans to grow its business globally. It moved its entire data platform into BigQuery (/bigquery) as its main data warehousing platform and relied on Cloud Storage (/storage) for storing documents. The ability of BigQuery to run real-time analytics makes it a great tool for segmentation and targeting of its user base, something that is crucial to the business as it looks to scale.

Cleartrip also required clear containerization and the ability to manage spikes in traffic at any time of day. Being in the travel business, it often sees sudden spikes in traffic, especially close to festivities and the holiday season, when people are typically

searching for flights and things to do in the places they intend to visit. When this happens, the company would be hard pressed to increase its capacity on short notice. Manoj says that this often ends up being a tedious process. "The high availability of infrastructure components on Google Kubernetes Engine (/kubernetes-engine) (GKE) means we can easily increase our capacity and decrease it when we need to."

Beyond that, being able to quickly scale infrastructure resources helps Cleartrip push out new initiatives quickly. "Our teams can focus on developing the app and can deploy and go live when ready, without having any dependency on infrastructure availability," says Manoj.

On top of its core business, Cleartrip provides white-label travel solutions by allowing large organizations to build their own solutions using its APIs. As an example, it offers medium and large businesses distribution APIs so that they can develop their own B2B and B2C offering. "Several partners we work with require global service-level agreements, so our ability to launch new solutions quickly is beneficial as we are able to provide a more predictable uptime for them."

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A collaborative partnership approach

There can be many misconceptions when migrating to the cloud, especially when moving for the first time. On top of that, teams needed to adapt to new ways of working, so one of the challenges for Manoj was to ensure that everyone was game to try new solutions. The move was an organization-wide change where everyone has a point of view so it was important to bring people through this cycle with a positive mindset.

Manoj finds working with Google Cloud very comforting because of the collaborative approach. "The Google Cloud team took the approach of educating us, rather than pressuring us to make a

sale. The process took a longer time, but it was a process that we all needed." He says that by doing so, everyone had a chance to familiarize themselves with the new ways of working and gradually appreciate and adapt to change.

"Of course, with great power comes great responsibility. The applications team now realize that the ball is completely in their court and they cannot blame anyone if something isn't right, but taking ownership is what we strongly believe in, and teams in general are very happy."

Cleartrip migrates to Google Cloud ...



